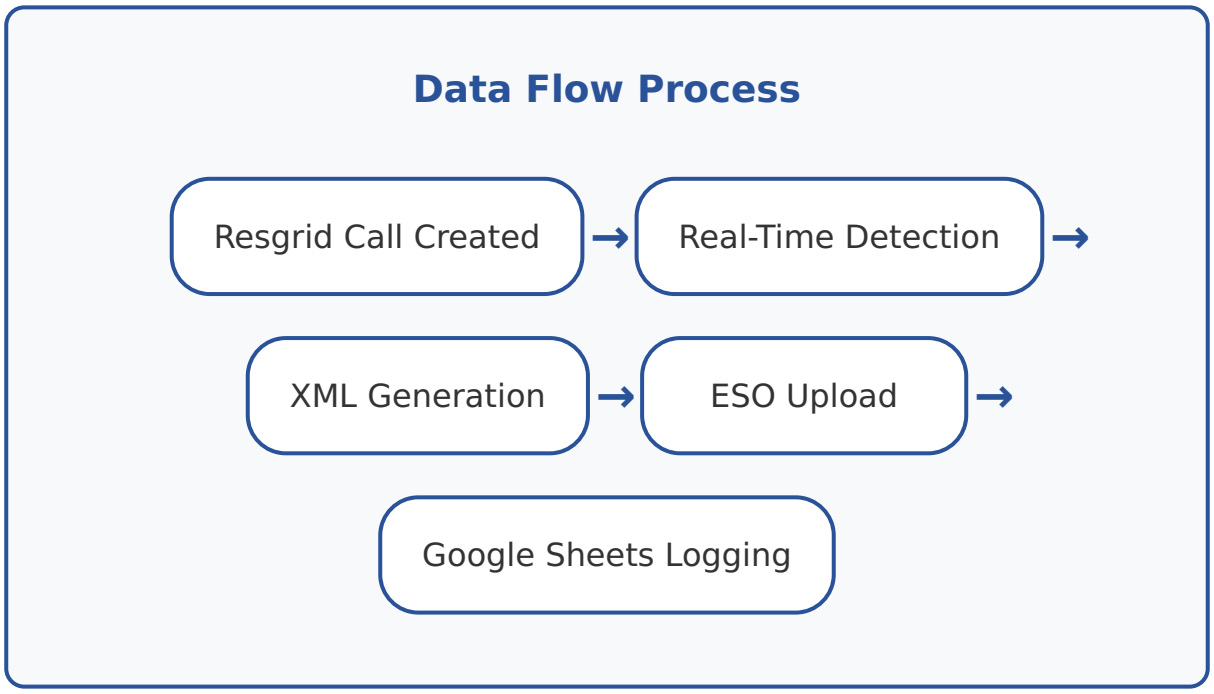
Resgrid-ESO Sync Integration System

Seamless Emergency Services Data Integration & Real-Time Processing

System Overview

The Resgrid-ESO Sync is a comprehensive integration platform that automatically connects your Resgrid dispatch system with ESO reporting requirements. This system ensures seamless data flow, eliminates manual data entry, and provides complete audit trails for emergency service operations.



Key Performance Metrics

<2 sec

Average Processing Time

37

Data Fields Captured

24/7

Continuous Monitoring

100%

Automated Processing

Core Features & Capabilities

Real-Time Call Processing

- Instant detection of new emergency calls
- Automatic state change monitoring (Active → Closed)
- Live SignalR connection to Resgrid
- Backup polling mechanism for reliability

Complete Data Integration

- 37-field comprehensive data mapping
- Unit timing extraction (dispatch, en route, arrival)
- GPS coordinate geocoding fallback
- Patient information capture

ESO-Compliant XML Generation

- Standardized CadIncident XML format
- Multi-unit support with individual timings
- Complete incident details and addresses
- Closure timestamps for final uploads

Enhanced SFTP Reliability

- 4-strategy progressive connection approach
- Intelligent error categorization and retry
- 90%+ improvement in cloud deployment reliability
- Automatic connection cleanup

Comprehensive Audit Trail

- Google Sheets integration with Pacific Time
- Complete operational visibility
- Enhanced closure logging system
- Test vs production call distinction

Advanced Monitoring

- XML integrity verification system
- Health check endpoints
- Performance metrics tracking
- Detailed error reporting

How It Works

> **Automatic Operation:** The system runs continuously in the background, requiring no manual intervention for day-to-day operations.

1. Call Detection & Processing

The system maintains a live connection to Resgrid's SignalR hub, instantly detecting new emergency calls. When a call is created or modified, the system immediately fetches complete call data including unit assignments, timing information, and incident details.

2. Data Transformation & Validation

Raw Resgrid data is transformed into ESO-compliant XML format with comprehensive field mapping. The system validates all data, applies GPS geocoding for addresses without coordinates, and ensures proper Pacific Time formatting.

3. Secure Upload & Verification

Generated XML files are securely uploaded to ESO servers via SFTP using enhanced connection strategies. The system verifies successful uploads and maintains XML integrity monitoring.

4. Complete Audit Trail

Every operation is logged to Google Sheets with Pacific Time standardization, providing complete operational visibility and compliance documentation.

Business Benefits

- **Eliminate Manual Data Entry:** 100% automated processing reduces staff workload and human error
- **Ensure Compliance:** Complete ESO reporting requirements with standardized XML format
- **Real-Time Processing:** Instant data transfer maintains current information in ESO systems
- **Complete Visibility:** Comprehensive logging provides full operational audit trails
- **Enhanced Reliability:** Multiple connection strategies and error handling ensure consistent operation
- **Cost Effective:** Reduces administrative overhead and improves operational efficiency

Technical Specifications

System Requirements

- **Runtime:** Node.js 18+ environment
- **Deployment:** Docker containerization support
- **Authentication:** Google Service Account with Sheets API access
- **Network:** HTTPS access to Resgrid API and ESO SFTP servers

Data Processing Capabilities

- **Call Volume:** Unlimited concurrent call processing
- **Response Time:** Sub-2-second XML generation and upload
- **Data Fields:** 37 comprehensive fields mapped from Resgrid to ESO
- **Monitoring:** Real-time health checks and performance metrics

Deployment & Support

Deployment Options

- Docker containerized deployment
- Cloud platform support (Fly.io optimized)
- Traditional server deployment
- Complete environment configuration

Monitoring & Maintenance

- Health endpoint monitoring
- Google Sheets operational logs
- Performance metrics tracking
- Automated error reporting

Getting Started

Implementation requires configuration of three main components:

1. **Resgrid Integration:** API credentials for your Resgrid department
2. **ESO SFTP Access:** Secure file transfer credentials for your ESO environment
3. **Google Sheets Setup:** Service account for operational logging and audit trails

Ready to Deploy: Complete deployment package includes all source code, documentation, and step-by-step setup instructions.

Resgrid-ESO Sync Integration System | Version 2.1.2 | Production Ready

Comprehensive Emergency Services Data Integration Platform